RUTGERS

Semantics II

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Please pick up your **name card** from the front

office hours: Mondays, 2:00-3:00 pm

Basement, Room 003, **Linguistics Building**

Previously on 201

- defining the meaning of words **by sense** and **by reference**:
  - a word's **sense** is the **mental concept** of the word's meaning or a dictionary definition of the word's meaning
  - a word's **reference** is the set of entities (referents) that the word **refers to**

- consider the word *university*
  - **sense**: an institution of higher education
  - **reference**: Rutgers, NYU, CUNY, Harvard, MIT, Penn, Stanford, Johns Hopkins...

- **Semantics** is the study of **literal** meaning in language
  - lexical semantics (words) and compositional semantics (sentences)

- lexical meaning is **conventional** and **arbitrary**
  - under this, we can have different relations: synonym, antonynm, hypernym, hyponym

- compositional meaning includes **truth values** of the sentence
  - sentences can have multiple propositions with different **truth values**

Entailment

ENTAILMENT

a proposition ***p*** entails another proposition ***q*** **if**, **whenever *p*** is **TRUE**, ***q*** must also be **TRUE**

more formally: ***p*** entails ***q*** **iff "whenever *p*** is **TRUE**, ***q*** is **TRUE**

iff: if and only if

"**whenever**": it means you cannot find any possible world, where **q** is **FALSE** while **p** is **TRUE**

Entailment





the relationship between the propositions in a sentence is called **entailment**

- a proposition ***p*** entails another proposition ***q*** **if**, **whenever *p*** is **TRUE**, ***q*** must also be **TRUE**

the sentence (1) *Earth is a rocky planet.* **entails** the sentence (2) *Earth is a planet.*

- for (1) *Earth is a rocky planet.* to be **TRUE**, (2) *Earth is a planet.* must also be **TRUE**

the sentence (3) *Snoopy is a beagle.* **entails** the sentence (4) *Snoopy is a dog.*

- for (3) *Snoopy is a beagle.* to be **TRUE**, (4) *Snoopy is a dog.* must also be **TRUE**

Possible Worlds

- one way to think about entailment is to imagine the **possible worlds** (alternative realities) in which a sentence would be true
- consider the sentence ***Merlin has a dog.***
- There are many possible worlds in which ***Merlin has a dog*** would be **TRUE**

World 1



World 2



World 3



World 4



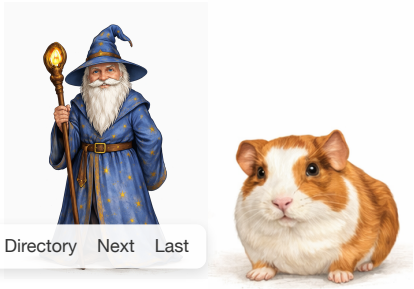
Possible Worlds

There are also many possible worlds in which ***Merlin has a dog*** would be **FALSE**

World 5



World 6



World 7



World 8



World 9



Possible Worlds

knowing the **meaning** of a sentence involves knowing in which possible worlds the sentence would be **TRUE**

World 1



World 2



World 5



World 6



World 8



World 9



Possible Worlds

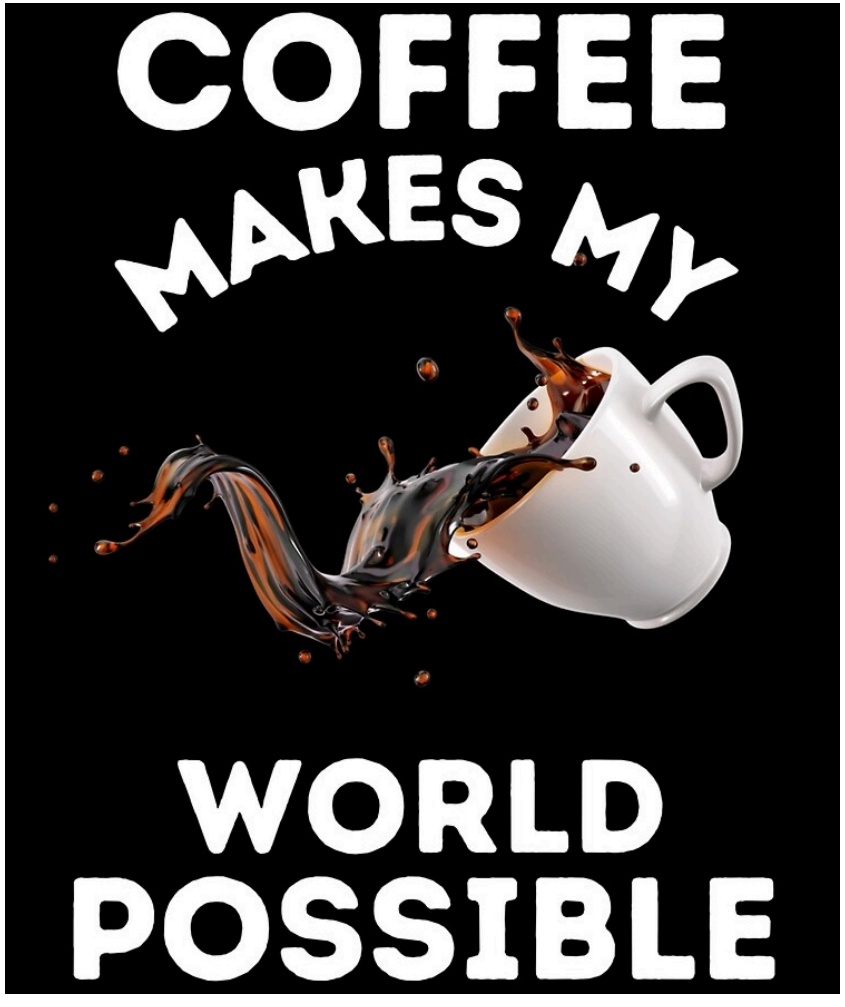
now consider the following sentences:

- can you think of a possible world in which only one of these sentences is **TRUE**?
- can you think of a possible world in which two or more of these sentences are **TRUE**?

(a) Merlin has a fluffy dog.

(b) Merlin has a dog.

(c) Merlin has a fluffy dog and a cat.



Entailment and Possible Words

- recall our definition of **entailment**
- (a) **entails** (b) because (a) cannot be TRUE while (b) is FALSE:
  - there is **NO** possible world in which *John has a fluffy dog but not a dog*
- (b) does **NOT entail** (a) because (b) can be TRUE while (a) is FALSE:
  - there is a possible world, in which *John could have a short-haired dog*
- (c) **entails** both (a) and (b) because (c) cannot be TRUE while (a) or (b) is FALSE

entailment: a proposition ***p*** entails another proposition ***q*** **if**, **whenever** ***p*** is TRUE, ***q*** must also be TRUE

in other words: ***p*** entails ***q*** when there is **NO** possible world in which ***p*** is TRUE and ***q*** is FALSE

(a) Merlin has a fluffy dog.

(b) Merlin has a dog.

(c) Merlin has a fluffy dog and a cat.

Entailment

consider the following sentences:

(a) Alicia drank coffee at midnight.

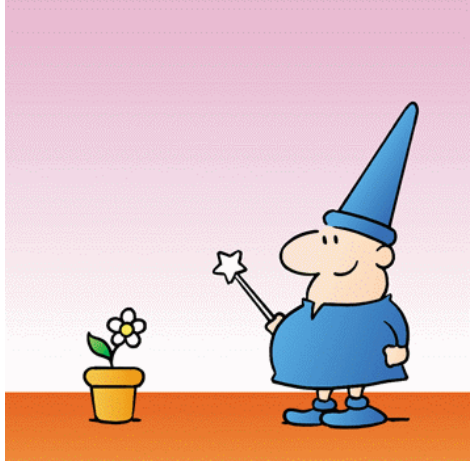
(b) Alicia drank coffee.

(c) Alicia drank coffee at midnight in the living room.

- does (a) **entail** (b)?
- does (a) **entail** (c)?
- does (b) **entail** (c)?
- does (c) **entail** (a)?

entailment: a proposition ***p*** entails another proposition ***q*** **if**, **whenever** ***p*** is TRUE, ***q*** must also be TRUE

in other words: ***p*** entails ***q*** when there is **NO** possible world in which ***p*** is TRUE and ***q*** is FALSE



Entailment

entailment: a proposition ***p*** entails another proposition ***q*** **if**, **whenever *p*** is TRUE, ***q*** must also be TRUE

in other words: ***p*** entails ***q*** when there is **NO** possible world in which ***p*** is TRUE and ***q*** is FALSE

consider the following sentences:

(a) Alicia drank coffee at midnight.

(b) Alicia drank coffee.

(c) Alicia drank coffee at midnight in the living room.

- does (a) **entail** (b)? *yes: if Alicia drank coffee at midnight is true, it must be true that Alicia drank coffee*
- does (a) **entail** (c)? *no: Suppose Alicia drank coffee at midnight in the kitchen. (a) will be true while (b) is false*
- does (b) **entail** (c)? *no: Suppose Alicia drank coffee in the morning. (b) will be true while (c) is false*
- does (c) **entail** (a)? *yes: If Alicia drank coffee at midnight in the living room is true, it must be true that Alicia drank coffee at midnight*

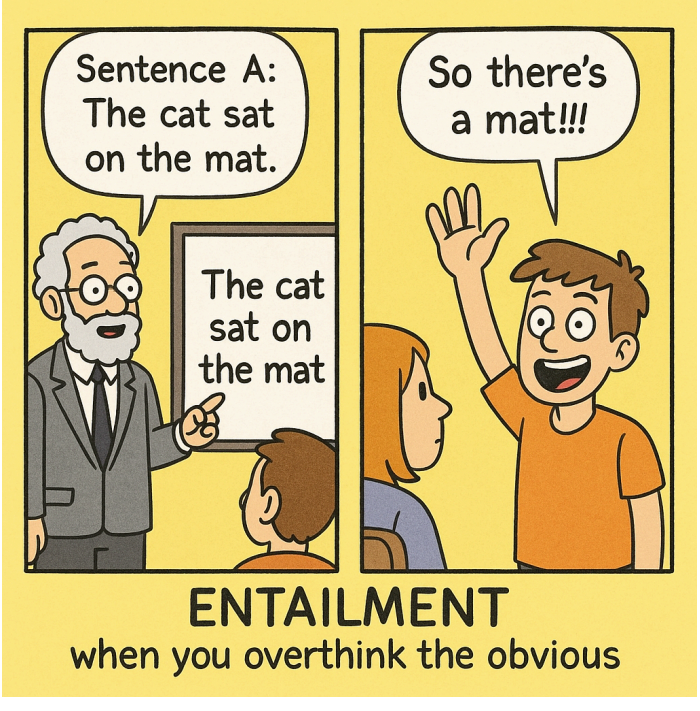
Logical Entailment

- **p** *entails* **q** for logical reasons, **regardless** of world knowledge or lexical meaning

  p: **The cat sat on the mat.**
  q: **There is a cat/mat.**

  p: **(a dog): I heard the can opener.**
  q: **(Therefore for the dog), FOOD EXISTS!.**

- We got the entailment relation between **p** and **q** by logical inferences



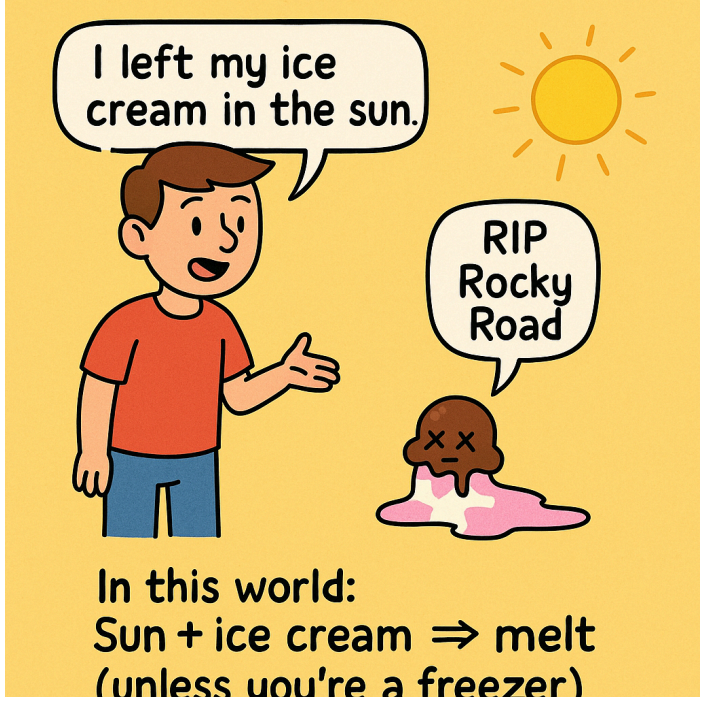
World-specific Entailment

- **p** *entails* **q** due to world knowledge or lexical meaning

  p: **Charles III is the King of the United Kingdom.**
  q: **Queen Elizabeth II was Charles's mother.**

  p: **Merlin is from Scotland.**
  q: **Merlin is from UK.**

- We got the entailment relation between **p** and **q** by making reference to how the world works, or to knowledge of lexical meaning

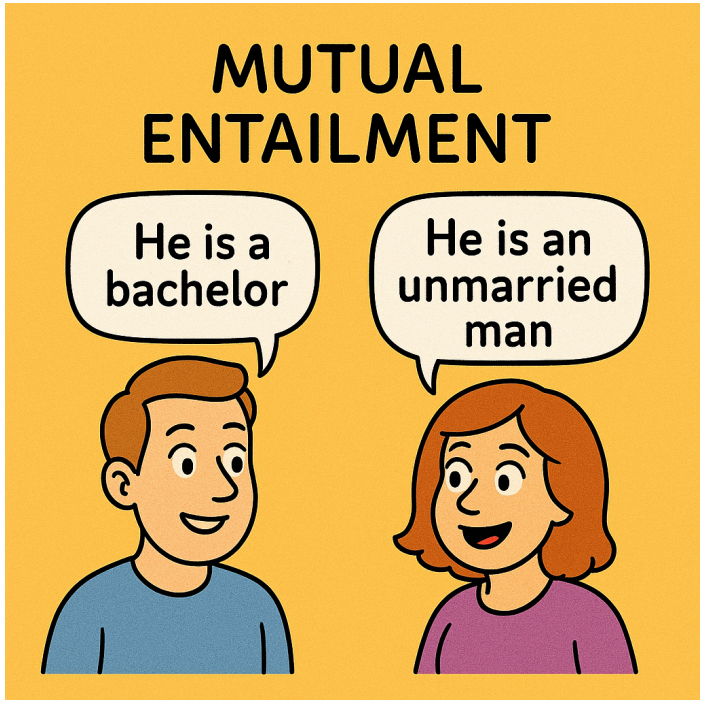


Mutual Entailment

- **p** *entails* **q** **AND** **q** *entails* **p** as well.

  p: **Merlin is a sorcerer.**
  q: **Merlin is a wizard.**

  p: **Merlin is immortal**
  q: **Merlin is not capable of dying.**



Exercise I: Entailment

entailment: Sentence **p** entails sentence **q** if
q is true whenever **p** is true.

For each pair of sentences, which one **entails** the other one?

- **p**. Charles III is the King of the United Kingdom.
  q. Queen Elizabeth II was Charles's mother.

- **p**. I am tired.
  q. I am very very very very very very tired.

- **p**. Bob live above a restaurant on Ocean Avenue.
  q. Bob and Linda lives above a restaurant on Ocean Avenue.

- **p**. The kids trick-or-treated on Halloween.
  q. The kids trick-or-treated in scary costumes on Halloween.



Exercise II: Entailment

entailment: Sentence **p** entails sentence **q** if
q is true whenever **p** is true.

work in **GROUP**, think of three sentences that are entailed by each of these sentences:

- **a**.
  The spotted lanternfly is an insect that has spread invasively to New Jersey and can cause damage to forests and fruit crops.
- **b**.
  The starship Enterprise explored strange new worlds, sought out new life and new civilizations, and boldly went where no one had gone before.
- **c**.
  Controversially, DeLa chose to eliminate herself from the competition after winning the lip-sync battle, which allowed Trixie, Kennedy, and Shangela to all stay in the competition.
- **d**.
  Mario defeated Bowser, but then all he found in the dungeon was Toad, who told him that the princess was in another castle.

entailment: Sentence **p** entails sentence **q** if
q is true whenever **p** is true.

Exercise III: Entailment Chain

consider the following sentences:

- (a) Merlin drank coffee.
- (b) Merlin drank coffee in the library.
- (c) Merlin drank coffee at midnight in the library.

1. does a **entail** b?
2. does b **entail** c?
3. does c **entail** b?
4. does b **entail** a?

what is the **difference** among these three sentences?

"i study better at the library or a coffee shop"

me at the library or coffee shop:



entailment: Sentence **p** entails sentence **q** if **q** is true whenever **p** is true.

Exercise III: Entailment Chain

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2. does b **entail** c?
3. does c **entail** b?
4. does b **entail** a?

5. does c **entail** a?

entailment: Sentence **p** entails sentence **q** if **q** is true whenever **p** is true.

Exercise III: Entailment Chain

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3. does c **entail** b?
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5. does c **entail** a?

TRANSITIVITY

a property where a **relation** transfers across elements

for instance, **if** A=B and B=C, **then** A=C

generally it refers to a "**passing over**" of a relationship, action, or property across elements

Entailment Notes I

- importantly, not all entailment relations can be identified so easily from the **syntactic structure** of the sentences and/or the **particular words** that they contain

- consider three pairs of sentences below:

  (a). *Arthur **managed to** leave on time*.
  (b). *Arthur left on time*.

  (c). *Arthur **tried to** leave on time*.
  (d). *Arthur left on time*.

  (e). *Arthur **attempted to** leave on time*.
  (f). *Arthur left on time*.

1. does (a) **entail** (b)?
2. does (c) **entail** (d)?
3. does (e) **entail** (f)?

Entailment Notes I

- importantly, not all entailment relations can be identified so easily from the **syntactic structure** of the sentences and/or the **particular words** that they contain

- consider three pairs of sentences below:

  (a). *Arthur **managed to** leave on time*.
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  (d). *Arthur left on time*.

  (e). *Arthur **attempted to** leave on time*.
  (f). *Arthur left on time*.

1. does (a) **entail** (b)?
2. does (c) **entail** (d)?
3. does (e) **entail** (f)?

in fact, syntactic structure is a poor indicator of entailment

afterall, entailment is a **semantic relation**, not a syntactic one

Entailment Notes II: 'Weird' Case

- there are sentences that are **ALWAYS** true and they are **true** in every possible world
    - (a) *A dog is a dog*.
    - (b) *I will either save the world or I will not save the world*.
- given our definition of entailment, such sentences are **entailed by all other sentences**
    - (c) *Merlin is a musical actor*.
- (a) and (b) are **always** true, so whenever (c) is true, they will be true:
    - if Merlin **IS** actually a musical actor in a possible world, then "*A dog is a dog*." is **true**
- we call this: entailment holds **vacuously**
    - why we call it "vacuously"?

Presupposition

PRESUPPOSITION

A SPECIAL TYPE OF ENTAILMENT

---

for two propositions p and q:

p presupposes q iff both p and the negation of p entail q

Presupposition: a Special Type of Entailment

- consider the following two pairs of sentences:

(a). *My brother Jeong Kwan lives in Korea*.

(b). *I have a brother*.

does (a) entail (b)?

(c). *My brother Jeong Kwan* ***does not*** *live in Korea*.

(b). *I have a brother*.

does (c) entail (b)?

Presupposition: a Special Type of Entailment

- consider the following two pairs of sentences:

(a). *My brother Jeong Kwan lives in Korea*.

(b). *I have a brother*.

does (a) entail (b)?

(c). *My brother Jeong Kwan* ***does not*** *live in Korea*.

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- this entailment **survives negation** so it is also a **presupposition**

Presupposition: a Special Type of Entailment

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does (a) entail (b)?

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does (c) entail (b)?

- this entailment **survives negation** so it is also a **presupposition**

- "if it is a **presupposition** relation, it must be an **entailment** relation as well"
- "if it is an **entailment** relation, it is **NOT necessarily** a **presuppostion** relation"

Presupposition: a Special Type of Entailment

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- this entailment **survives negation** so it is also a **presupposition**

- "if it is a **presupposition** relation, it must be an **entailment** relation as well"
- "if it is an **entailment** relation, it is **NOT necessarily** a **presuppostion** relation"

we also say that a **presuppostion** is the belief or background (tacit) assumption whose truth is taken for granted in a sentence or discourse

Presupposition: a Special Type of Entailment

- **FOR INSTANCE**:

(a). *Merlin drank coffee in the library*.

(b). *Merlin drank coffee*.

(c). *Merlin DID NOT drink coffee in the library*.

(b). *Merlin drank coffee*.

Presupposition: a Special Type of Entailment

- **FOR INSTANCE**:

(a). *Merlin drank coffee in the library*.

(b). *Merlin drank coffee*.

(a) entails (b)

(c). *Merlin DID NOT drink coffee in the library*.

(b). *Merlin drank coffee*.

(c) does NOT entail (b)

- therefore, (a) is NOT a **presupposition** of (b)

Presupposition: a Special Type of Entailment

- **FOR INSTANCE**:

(a). *Merlin drank coffee in the library*.

(b). *Merlin drank coffee*.

(a) entails (b)

(c). *Merlin DID NOT drink coffee in the library*.

(b). *Merlin drank coffee*.

(c) does NOT entail (b)

- therefore, (a) is NOT a **presupposition** of (b)

- note here: we are deciding the presupposition relation between **proposition (a) and (b)**
- however, given our definition of presupposition, we have to refer to another proposition, **c**, which is the **negation of b,** to determine presupposition relation

Exercise IV: Presupposition

presupposition: for two propositions **p** and **q**, **p** presupposes **p** iff **both p** and the **negation** of **p** **entail q**.

entailment: Sentence **p** entails sentence **q** **iff** **q** is true whenever **p** is true.

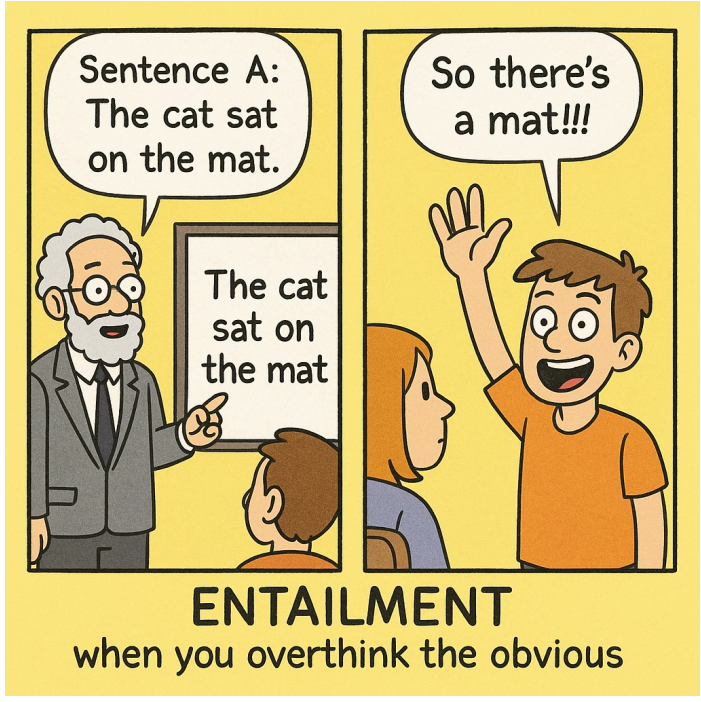
do following pairs of sentence fall under **presupposition** relation?

(1)
p: **The cat sat on the mat.**
q: **There is a cat/mat.**

(2)
p: **(a dog): I heard the can opener.**
q: **(Therefore for the dog), FOOD EXISTS!.**

(3)
p: **The king of the United States is bald.**
q: **There is a king of the United States.**

(4)
p: **Arthur is driving a blue car.**
q: **Arthur is driving a car.**



we will come back to **presupposition** again in **Pragmatics**

in that section, we will introduce different types of presupposition

Assignment IV will be published today (**Oct 21**)
and it is due **Oct 30** at **11:59 pm**

continue reading: the chapter **Semantics I: Compositionality**
from Victoria Fromkin et al.'s *An Introduction to Language*



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